Kartik Pal

Founding Team · Data Scientist · AI Engineer · LLM & RAG Systems · MLOps · pal28kartik67@gmail.com • +91 9711660529 •
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PROFESSIONAL SUMMARY

Founding team member at Vgenomics, where I built AI systems from the ground up, directly enabling ₹95 Lakh (₹75 Lakh from the Technology Development Board, Government of India, and ₹20 Lakh from IIM Bangalore) in external funding and supporting early-stage fundraising efforts. Data Scientist and AI Engineer with 2+ years of experience delivering production-grade ML, MLOps, and LLM-powered systems for high-stakes, data-intensive environments. Core expertise spans retrieval-augmented generation (RAG) pipelines, LangGraph agentic AI, and natural language processing (NLP). Designed event-driven MLOps architectures (MLflow, CI/CD, automated retraining) on AWS and GCP for scalable and robust deployment. Published researcher with peer-reviewed work in advanced NLP and complex data extraction, including genomic variant interpretation and rare disease diagnostics.

CORE COMPETENCIES

LLM & GenAI	RAG Pipelines · LangChain · LangGraph · Agentic AI
ML / DL	PyTorch · PyTorch Geometric · Graph Neural Networks (GNN) · Random Forest · XGBoost · Biomedical Knowledge Graphs · NLP
MLOps	MLflow · CI/CD · Automated Retraining · Model Registry · Event-driven Pipelines · Docker · Serverless Architecture
Cloud – AWS	Lambda · S3 · Batch · Athena · EC2 · ECR · ECS · RDS · API Gateway · NAT Gateway · CloudWatch · GuardDuty · IAM
Cloud – GCP	Cloud Run · Cloud Functions · Cloud Storage · Compute Engine · Artifact Registry · Cloud Scheduler · Cloud SQL
Backend & Data	Python · FastAPI · GraphQL · REST APIs · PostgreSQL · MySQL · S3 Data Lakes
Healthcare AI	HPO Ontology · ICD-10 · HL7 FHIR · HIPAA · EHR Processing · ACMG Variant Classification · Genomics

PROFESSIONAL EXPERIENCE

Data Scientist — Founding Team | Vgenomics | Noida, India | Apr *2024 – Present*

VUSPredict — ML-Based ACMG Variant Classification & MLOps

- Built ensemble ML pipeline (Random Forest, XGBoost + VEP/dbNSFP annotations) to classify Variants of Uncertain Significance (VUS) with probabilistic ACMG pathogenicity scoring and strong ROC-AUC performance.
- Designed event-driven MLOps on GCP (Cloud Functions, Cloud Run, FastAPI, MLflow) for experiment tracking, model registry, automated retraining, and continuous deployment as a serverless endpoint.

TxGNN — GNN-Based Drug Repurposing for Rare Diseases

- Adapted TxGNN—a Graph Neural Network foundation model—to predict zero-shot therapeutic candidates for rare diseases using heterogeneous biomedical knowledge graphs (disease–gene–drug) with PyTorch Geometric; enabled explainable multi-hop reasoning for treatment discovery.

Genomic Analysis Scalable Infrastructure — Serverless AWS Clinical Platform

- Integrated Dockerized genomic pipelines into the clinical UI via serverless AWS (Lambda → Batch → concurrent Docker containers on S3 upload triggers); secured with NAT Gateway, CloudWatch, and GuardDuty for clinical-grade compliance.

Genomic Variant Explanation Agent — Multi-Source Agentic AI

- Architected an agentic system synthesizing evidence from 15+ GB of genomic databases (ClinVar, gnomAD, OMIM via AWS Athena) to produce clinician-ready, per-claim citation-grounded variant reports—minimizing hallucination risk in clinical use. Deployed on AWS Lambda (serverless, auto-scaling).

RarePredict — AI Rare Disease Expert System

- Led development of a network-based diagnostic model, achieving 91% Top-5 accuracy on real-world patient data for rare disease prediction from symptom profiles.

- Built an end-to-end NLP + LLM RAG pipeline for HPO term extraction from free-text EHR notes (context-aware chunking) and an EHR Symptom Extractor using Docling for PDF parsing—both deployed as FastAPI services on GCP Cloud Run.

Data Science Intern | **Vgenomics** | Noida, India | Oct *2023 – Apr 2024*

- Developed the foundational rare disease prediction algorithm from symptom profiles—the core that became RarePredict. Built a PubMed literature mining tool to retrieve and index research papers by keyword.

RESEARCH & PUBLICATIONS

Rarepheno—AI-Driven Symptom Extraction (Under Review)

- NLP pipeline integrating DeepSeek, BioClinicalBERT, and HPO knowledge graphs; precision 0.830—outperforming PhenoGPT, ClinPhen, and MetaMap on clinical narrative extraction.

RareGEM—RAG-Based Gemini Framework for Rare Disease Phenotyping (Under Review)

- Gemini RAG framework with zero-shot, few-shot, and CoT prompting + semantic reranking; precision 0.862 on BioLarkGSC+ benchmark.

Book Chapter in Elsevier: AI-driven rare disease diagnosis—Generative AI, federated learning, and multimodal healthcare data fusion for precision diagnostics. [Link]

EDUCATION

MTech in Data Science — CGPA: 8.38 (2023–2025)
The NorthCap University, Gurgaon

BTech in Biotechnology — CGPA: 9.42 (2018–2022)
Amity University, Gurgaon